Multimodal Sound Localization and Visualization

Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of

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in
Computer Science and Engineering

Submitted by

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Declaration

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Abstract

This work develops a real-time acoustic camera that overlays microphone-array Direction-of-Arrival (DOA) heatmaps onto a live video feed. We implemented an Incoherent Broadband MUltiple SIgnal Classification or MUSIC-based localization pipeline and integrated it with a USB webcam to create a real-time acoustic imaging system suitable for speech and other broadband sources. The system was deployed on a MiniDSP UMA-16 microphone array with several optimizations for covariance estimation, eigen-decomposition, and real-time rendering. Multimodal extensions such as Eulerian Video Magnification (EVM) were explored but not included in the final system due to hardware constraints. We document the complete pipeline, hardware integration, evaluation on LOCATA and live recordings, and outline practical future directions.

1. Introduction

1.1 Overview and Motivation

Sound localization is the process of determining the direction or relative position of a sound source. In technical applications, this is typically achieved using arrays of microphones. By analyzing the differences in the time and phase of the sound arriving at each microphone, the system can calculate the source's location.

Acoustic cameras are a direct application of this technology. They consist of a microphone array and a video camera, where the calculated sound source location is visually overlaid as a heatmap on the video feed. This provides an intuitive way to "see" sound, which is invaluable for tasks like noise, vibration, and harshness (NVH) testing in the automotive industry, or for identifying faulty machinery in industrial settings. The core motivation for this project is the need for a reliable and precise method to pinpoint noise sources in these complex environments.

1.2 Related Work and Proposed Idea

Commercial acoustic cameras rely primarily on microphone-array processing to estimate sound source directions. While these systems work well, their performance can degrade in noisy or reverberant environments, and they generally do not exploit information available from the visual domain. Recent research in audio–visual perception suggests that combining acoustic cues with video-based motion analysis may improve robustness by providing additional spatial or temporal context.

Building on this motivation, our project focuses on developing a reliable real-time DOA estimation algorithm and integrating it with a live video feed for visualization. Multimodal extensions were considered during early design discussions, but the final system centers on accurate and efficient audio-based localization, leaving deeper audio–video fusion as future work.

2. Methodology

2.1 System Overview

The proposed system is implemented as a multi-threaded Python application composed of three distinct functional blocks: Data Acquisition, Signal Processing, and Visualization. To ensure real-time performance, audio capture and buffering are handled in a dedicated callback thread or the producer, while video capture, MUSIC processing, and rendering occur in the main execution loop or the consumer. The data flow relies on a First-In-First-Out (FIFO) queue structure to synchronize high-speed audio samples (f=48 kHz) with lower-framerate (30fps) video data. This decouples audio processing from camera I/O, preventing blocking when either subsystem lags.

2.2 Hardware Setup

- Microphone Array: MiniDSP UMA-16 (4x4 MEMS array) arranged in a Uniform Rectangular Array (URA) with an approximate spacing of 4.2cm.

- Optical Sensor: HP Webcam 1300 (640x480@30fps) with a horizontal Field of View (FOV) of 50°.

- Mounting and Fixturing: A custom mounting fixture was fabricated using 3D printing (PLA). This adapter interfaces a standard ¼”-20 UNC tripod thread with the specific mounting geometry of the UMA-16 PCB.

2.3 Signal Processing Pipeline

The core processing pipeline operates in frames of 4096 samples (~85ms) with a 50% overlap (hop size of 2048 samples). The processing at each frame is explained below:

2.3.1 Preprocessing

Each audio frame undergoes Hanning to reduce spectral leakage, followed by a Short-Time Fourier Transform (STFT) over each channel to convert signals into the frequency domain. The resulting spectrum is then divided into multiple frequency bands.

2.3.2 Cross-Spectral Matrix (CSM)

For every frequency band, the CSM is computed. Each CSM is a 16×16 complex-valued matrix encoding microphone correlations at that frequency band.

$$R_{inst}(f) = X(f)X^H(f)$$

Where $X(f)$ denotes the STFT values for all microphones within the frequency band f.

To ensure temporal stability and reduce jitter in the visualization, an exponential moving average is applied to the CSM:

$$\mathrm{R_t}(f) = (1 - \alpha)\mathrm{R_{t-1}}(f) + \alpha R_{inst}(f)$$

Where R_{inst} is the CSM computed from the current frame, and α is the smoothing factor (optimized to 0.3).

2.3.3 Subspace Decomposition

For each frequency band, the smoothed CSM R_t is eigendecomposed. The smallest values correspond to noise components. Their eigenvectors are used to form the noise subspace.

The largest values are used to form the signal subspace.

2.3.4 Music Spectrum Computation

For each candidate azimuth-elevation angle pair, a precomputed steering vector $a(f, \theta)$ is projected onto the noise subspace.

$$\mathrm{P(f,\theta)} = \frac{1}{\left\| E_{noise}^{H} a(f,\theta) \right\|}$$

Since the true DOA lies mostly in the signal subspace, and the noise subspace is orthogonal to the signal subspace, this yields a spatial spectrum on the azimuth-elevation grid with sharp peaks at the true source direction.

2.3.5 Broadband Aggregation

The spatial spectra from each band are normalized and summed to produce an incoherent broadband DOA estimate.

$$\mathrm{P_{total(\theta)}} = \sum_{\mathrm{f \in freqs}} \frac{\mathrm{P(f,\theta)}}{\max_{\theta'}\left(\mathrm{P}(f,\theta')\right)}$$

This improves robustness for practical applications like speech and other multi-frequency sources. We used 6 bands each of width 500Hz each between 500-3000Hz.

2.4 Optimization Strategies

To achieve real-time performance in the acoustic camera pipeline, several computational and algorithmic optimizations were implemented:

- Precomputation of Steering Matrices – Steering vectors for all frequency bands are computed once during initialization and reused for every frame. This removes repeated trigonometric function calls.
- Exponential CSM Smoothing – The recursive update rule eliminates the need to maintain a history of the CSM.
- Vectorized Linear Algebra – Covariance computation and projections are implemented using optimized NumPy vector operations.
- Noise-Gated Processing – Beamforming is skipped entirely for frames where the noise level doesn't reach a certain threshold (we have used 15dB).

3. Results

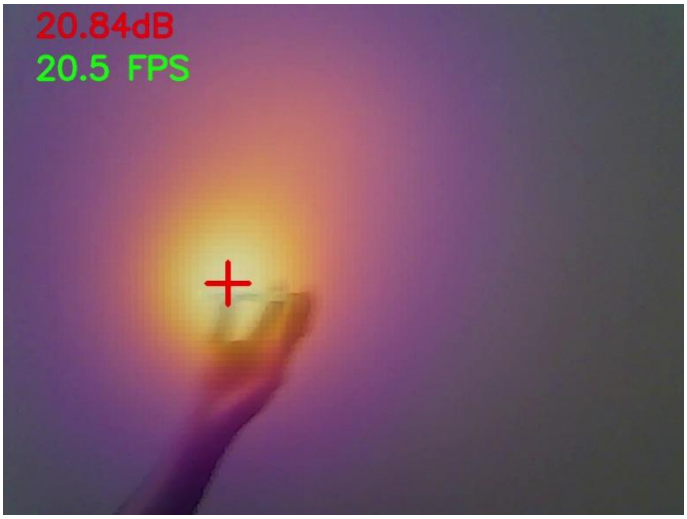
Since the system was deployed on real hardware without a motion-capture setup for ground-truth angles, the evaluation focuses on visual correctness, stability of the DOA estimates, and practical observations observed during live testing.

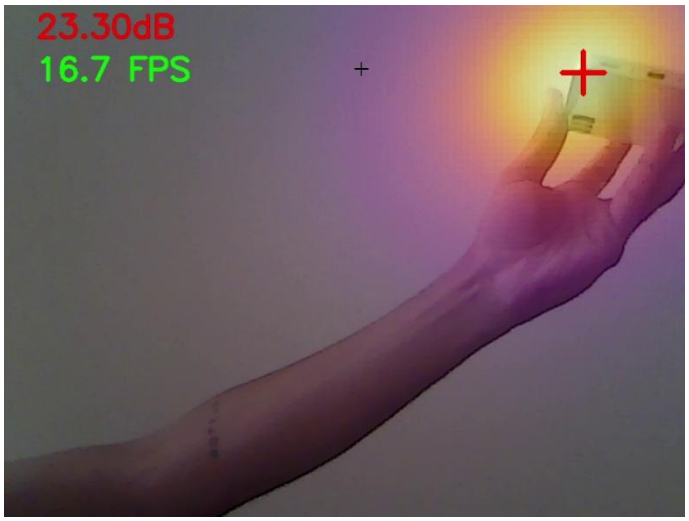
3.1 Performance Metrics

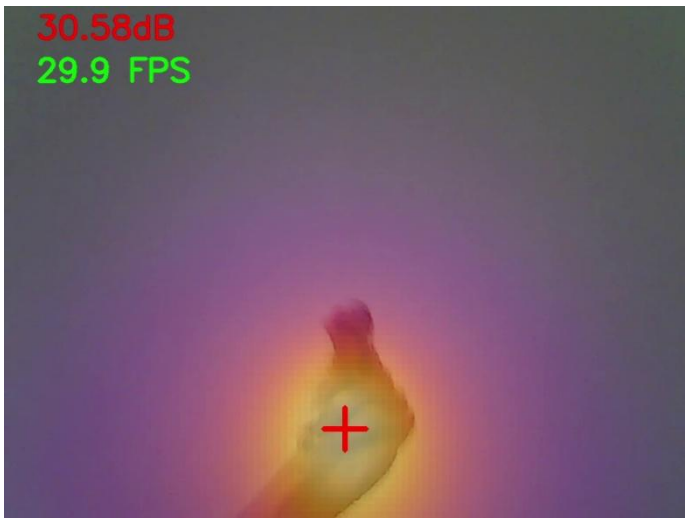
With an FOV of 50°x38°, we consistently achieved a minimum of 17-22fps when running the system with a grid resolution of 0.5° on an Intel Core Ultra 5 125H processor.

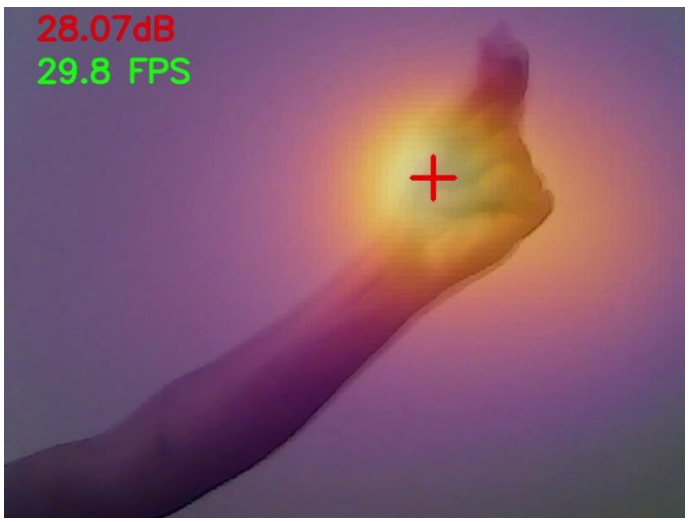
3.2 Qualitative DOA Visualization

The system successfully overlays the MUSIC-based DOA heatmap onto the live video feed in real time. Across multiple test scenarios such as speech, finger snaps, and phone audio playback the system produced consistent bright peaks in the heatmap corresponding to the true direction of the sound source. Moving the source smoothly resulted in the DOA peak tracking motion with minimal lag.









4. Future Work

Future extensions of this work can move in several practical and algorithmic directions. A key limitation of the current system is the absence of quantitative ground-truth evaluation, which can be addressed by using a controlled angular measurement setup to compute DOA error across distances and frequency bands. On the algorithmic side, implementing coherent broadband MUSIC could exploit phase continuity across frequency bands and improve robustness in reverberant environments. Real-time performance can also be enhanced by GPU acceleration, or a hierarchical algorithm where it adaptively moves from a coarse to fine grid. Finally, although visual micro-vibration methods such as EVM were not feasible under the current hardware constraints, future work may explore alternative multimodal fusion methods such as leveraging optical flow or higher-frame-rate cameras to combine visual cues with audio DOA estimates for more stable and reliable acoustic localization.

5. References

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